

Machine Learning - Undergraduate Level Assessment 12

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

As of 2020, which architecture is best for classifying high-resolution images?

- A. convolutional networks
- B. graph networks
- C. fully connected networks
- D. RBF networks

Question 2

(2 marks)

Which of the following tasks can be best solved using Clustering.

- A. Predicting the amount of rainfall based on various cues
- B. Detecting fraudulent credit card transactions
- C. Training a robot to solve a maze
- D. All of the above

Question 3

(2 marks)

Which one of the following is equal to $P(A, B, C)$ given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?

- A. $P(A | B) * P(B | C) * P(C | A)$
- B. $P(C | A, B) * P(A) * P(B)$

C. $P(A, B | C) * P(C)$

D. $P(A | B, C) * P(B | A, C) * P(C | A, B)$

Question 4

(2 marks)

A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?

A. $P(A|B)$ decreases

B. $P(B|A)$ decreases

C. $P(B)$ decreases

D. All of above

Question 5

(2 marks)

Which of the following are the spatial clustering algorithms?

A. Partitioning based clustering

B. K-means clustering

C. Grid based clustering

D. All of the above

True / False

Question 1

(2 marks)

K-fold cross-validation's computational complexity is quadratic in K, which increases significantly with more folds.

True

False

Question 2

(2 marks)

Transforming the data to a unit variance is sufficient to achieve the same projection as SVD in PCA.
Transforming the data to a zero median is sufficient to achieve the same projection as SVD in PCA.

True

False

Question 3

(2 marks)

Out-of-distribution detection is synonymous with background detection in machine learning.

True

False

Question 4

(2 marks)

The dimensionality of the null space of the matrix A is 0.

True

False

Question 5

(2 marks)

Applying an L 1 norm regularization penalty can result in some coefficients being zeroed out in linear regression.

True

False

Long Form

Question 1

(10 marks)

Write a function to remove characters from the first string which are present in the second string.

Type your answer here...



Question 2

(10 marks)

Write a function to divide two lists using map and lambda function.

Type your answer here...



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